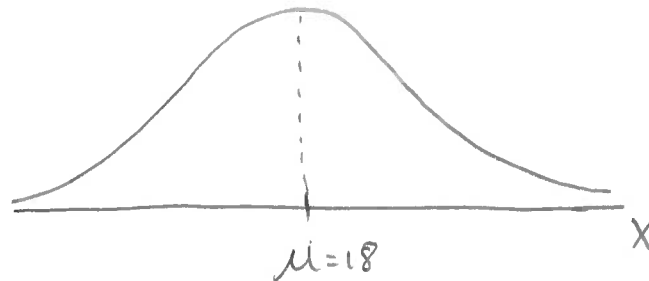


Example: Normal probabilities

①

Let $X \sim N(\mu=18, \sigma^2=25)$, thus $\sigma=5$



Calculate:

(a) $P(X < 18.6)$

(b) $P(X > 18.6)$

(c) $P(18 < X < 18.6)$

(d) $P(17.4 < X < 18)$

(e) $P(X < 18)$

(f) $P(X > 17 \cap X < 19)$

(g) $P(X < 19.2 \mid X > 17.3)$

(h) $P(X = 17)$

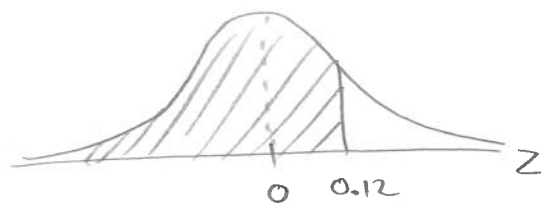
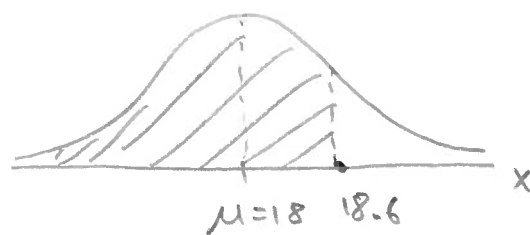
(a)

$$P(X < 18.6)$$

$$= P\left(\frac{X - \mu}{\sigma} < \frac{18.6 - 18}{5}\right)$$

$$= P(Z < 0.12)$$

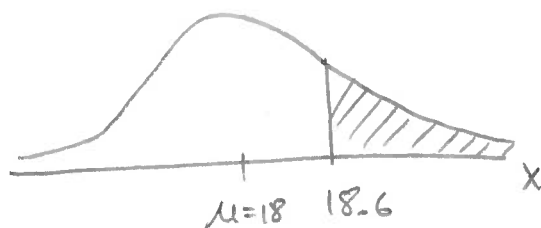
$$= \underline{0.5478} \rightarrow$$



$$(b) P(X > 18.6)$$

$$= 1 - P(X < 18.6)$$

$$= 1 - 0.5478 = \underline{0.4522} \rightarrow$$

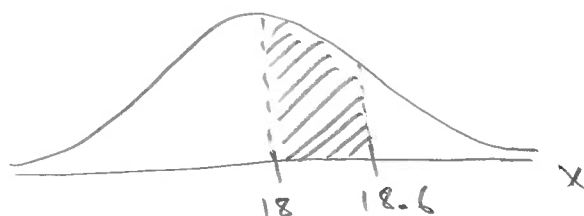


$$(c) P(18 < X < 18.6)$$

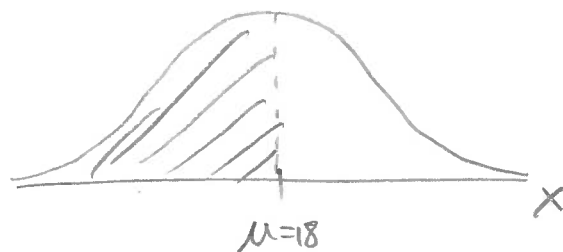
$$= P(X < 18.6) - P(X < 18)$$

$$= P(Z < 0.12) - P(Z < 0)$$

$$= 0.5478 - 0.5 = \underline{0.0478} \rightarrow$$



$$(e) P(X < 18) = \underline{0.5} \rightarrow$$



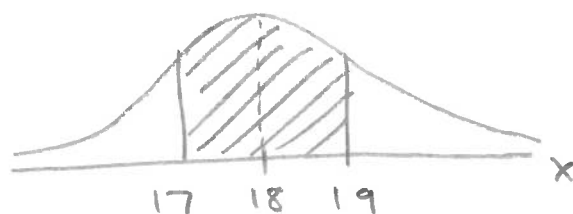
$$(f) P(X > 17 \cap X < 19)$$

$$= P(17 < X < 19)$$

$$= P(X < 19) - P(X < 17)$$

$$= P(Z < 0.2) - P(Z < -0.2)$$

$$= 0.5793 - 0.4207 = \underline{0.1585} \rightarrow$$



$$(g) P(X < 19.2 | X > 17.3) = \frac{P(X < 19.2 \cap X > 17.3)}{P(X > 17.3)}$$

$$= \frac{P(17.3 < X < 19.2)}{1 - P(X < 17.3)}$$

$$= \frac{P(X < 19.2) - P(X < 17.3)}{1 - P(X < 17.3)}$$

$$= \frac{P(Z < 0.24) - P(Z < -0.14)}{1 - P(Z < -0.14)}$$

$$= \frac{0.5948 - 0.4443}{1 - 0.4443} = \underline{0.2708} \rightarrow$$

$$(h) P(X = 17) = \underline{0} \rightarrow$$